EOQ Practice Worksheet

Hochschule Heilbronn – Business Administration
Topic: Procurement and Material Management

Instructions: Calculate the Economic Order Quantity (EOQ) for each question below. Use the formula:
EOQ = √((2DS)/H)
where D = Annual demand, S = Ordering cost, and H = Holding cost per unit per year.

Part A – Student Worksheet

Question 1 – Basic EOQ Calculation

Given:
- Annual demand (D) = 12,000 units
- Ordering cost (S) = €60 per order
- Holding cost (H) = €1.50 per unit per year

Find: EOQ

Answer: _____

Question 2 – EOQ, Orders per Year, and Time Between Orders

Given:
- Annual demand (D) = 24,000 units
- Ordering cost (S) = €40 per order
- Holding cost (H) = €2 per unit per year

Find:
1. EOQ
2. Number of orders per year
3. Time between orders (weeks)

Answers:

1. EOQ = _____
2. Orders per year = _____
3. Time between orders = _____

Question 3 – EOQ with Higher Holding Cost

Given:
- Annual demand (D) = 15,000 units
- Ordering cost (S) = €75 per order
- Holding cost (H) = €3 per unit per year

Find: EOQ

Answer: _____

Question 4 – EOQ and Total Annual Cost

Given:
- Annual demand (D) = 10,000 units
- Ordering cost (S) = €50 per order
- Holding cost (H) = €2 per unit per year
- Unit cost (C) = €20

Find:
1. EOQ
2. Orders per year
3. Total Annual Cost

Answers:
1. EOQ = _____
2. Orders per year = _____
3. Total Cost = _____